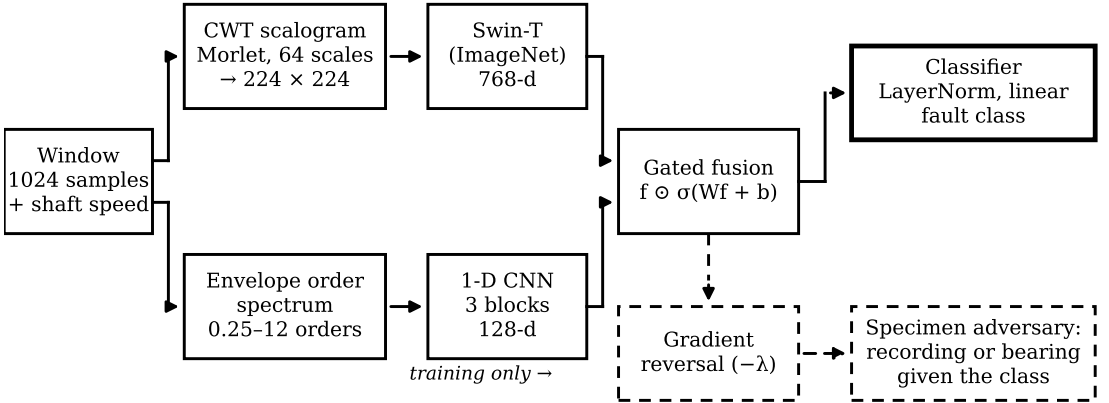
Window
1024 samples
+ shaft speed
CWT scalogram
Morlet, 64 scales
→ 224 × 224
Swin-T
(ImageNet)
768-d
Envelope order
spectrum
0.25–12 orders
1-D CNN
3 blocks
128-d
training only →
Gated fusion
f ⊙ σ(Wf + b)
Classifier
LayerNorm, linear
fault class
Gradient
reversal (−λ)
Specimen adversary:
recording or bearing
given the class